

Translations of sine and cosine functions



1

a $y = -1 + \sin x$

b $y = -3 + \cos x$

2

a Horizontal translation of 135° left.

b 135°

3

$y = \sin(x - 30^\circ)$

4

$y = \cos\left(x - \frac{\pi}{6}\right)$

5

a $y = \sin(x + 60^\circ)$

b 1

c 360°

6

a $y = \cos\left(x + \frac{\pi}{3}\right)$

b 1

c 2π

7

$-90^\circ, 270^\circ$

8

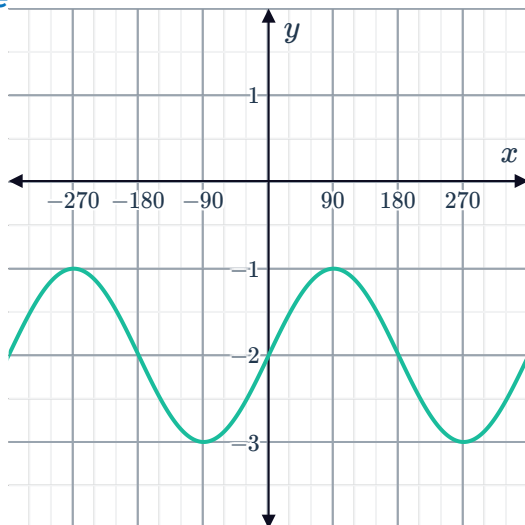
a 360°

b 1

c -1

d -3

e



9

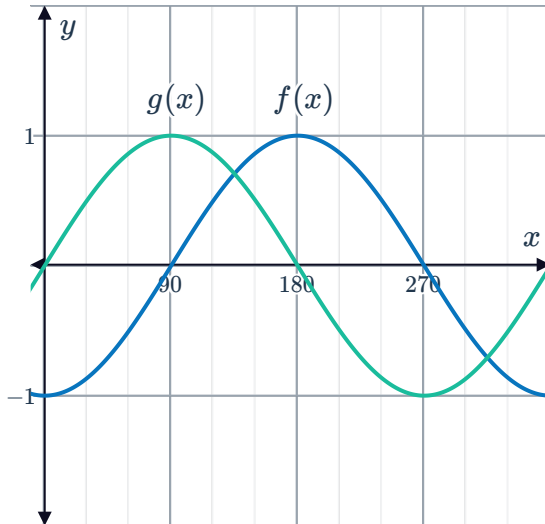


a

x	0	90°	180°	270°	360°
$f(x)$	0	1	0	-1	0
$g(x)$	-1	0	1	0	-1

b Horizontal translation 90° to the right

c



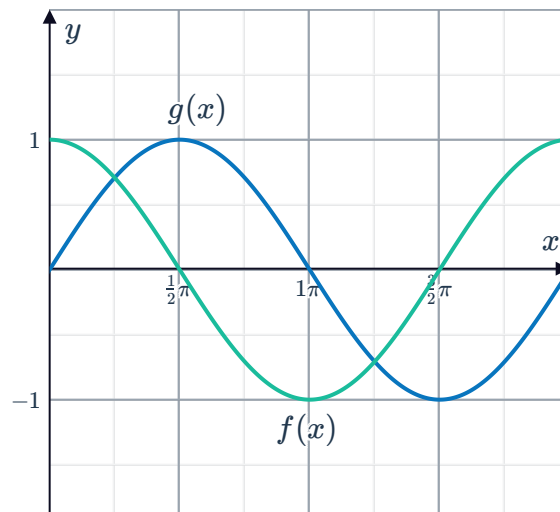
10

a

x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$f(x)$	1	0	-1	0	1
$g(x)$	0	1	0	-1	0

b Horizontal translation $\frac{\pi}{2}$ units to the right

c





11

a

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\cos x$	1	0.71	0	-1	-0.71	0	0.71	1
$\cos\left(x + \frac{\pi}{4}\right)$	0.71	0	-0.71	-0.71	0	0.71	1	0.71

b To the left

Dilations of sine and cosine functions

12 $y = 2 \cos x$

13 $y = 2 \sin x$

14 $y = \cos\left(\frac{1}{2}x\right)$

15

a Yes

b Yes

c No

d No

e No

f Yes

g No

h No

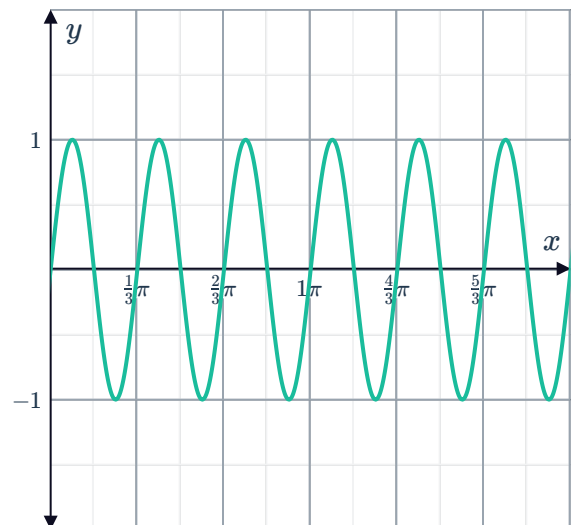
i Yes

j No

16

a $y = 3 \sin 6x$

b





17

a $y = \sin 2x$

b $y = \cos 4x$

18

a 4

b $\frac{1}{9}$

19

a No

b Yes

c No

d No

20

a 5

b -5

c 5

21

a 360°

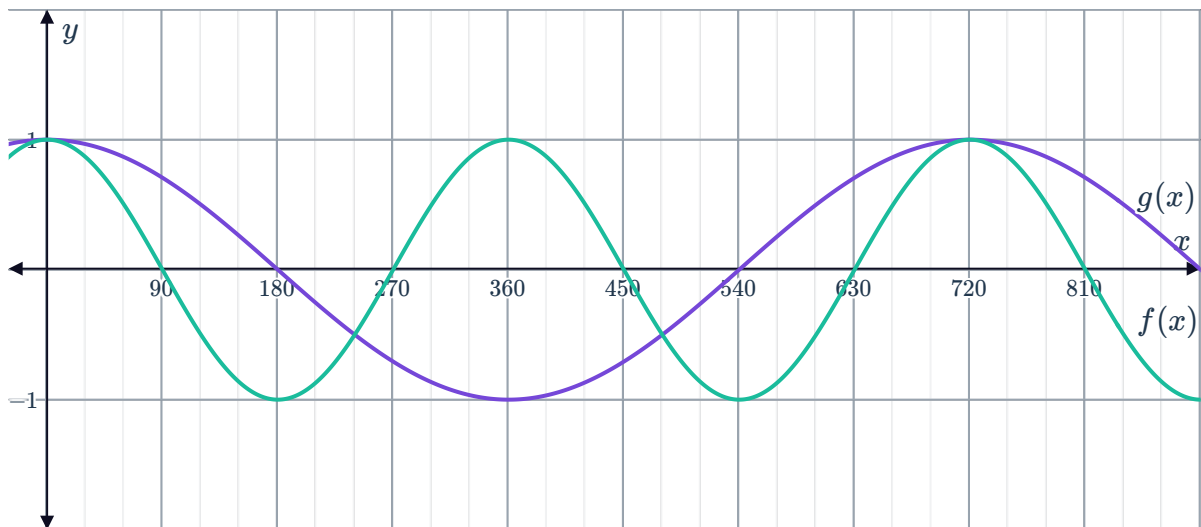
b

x	0°	180°	360°	540°	720°	900°	1080°
$g(x)$	1	0	-1	0	1	0	-1

c 720°

d Horizontal stretch by a factor of 2

e



f No



Series of transformations

22 The graph of $y = \cos x$ undergoes a phase shift of 45° to the left, and vertically translated downwards by 2 units.

23

a Horizontal stretch by a factor 2 and a horizontal translation (phase shift) of 30° to the left.

b $g(x) = \sin(2x + 60^\circ)$

24

a Horizontal stretch by a factor 3 and a horizontal translation (phase shift) of $\frac{\pi}{9}$ units left.

b $g(x) = \cos\left(3x + \frac{\pi}{3}\right)$ or $g(x) = \cos\left(3\left(x + \frac{\pi}{9}\right)\right)$

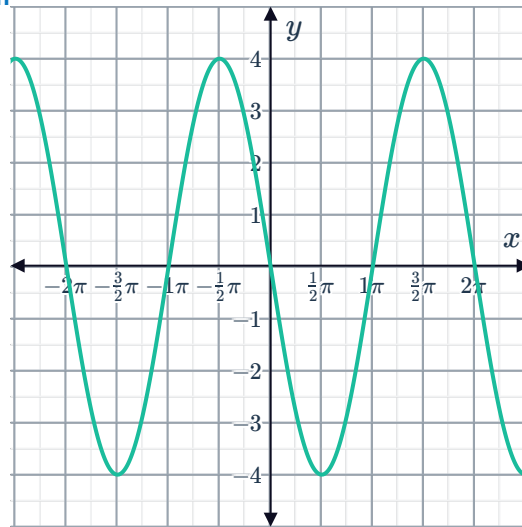
25

a

i 4

ii π

iii

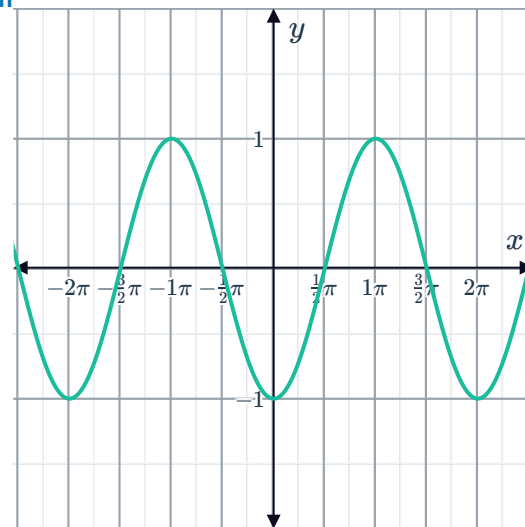


b

i 1

ii $\frac{\pi}{2}$ 

iii

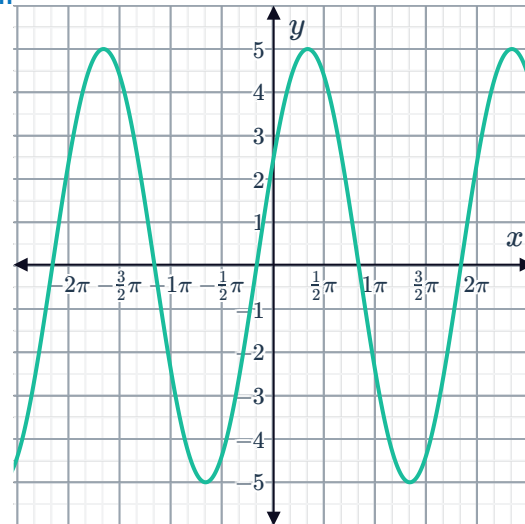


c

i 5

ii $\frac{\pi}{3}$

iii

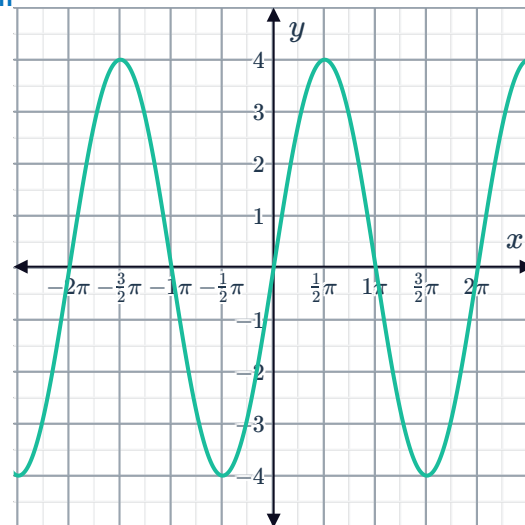


d

i 4

ii $-\frac{\pi}{2}$

iii

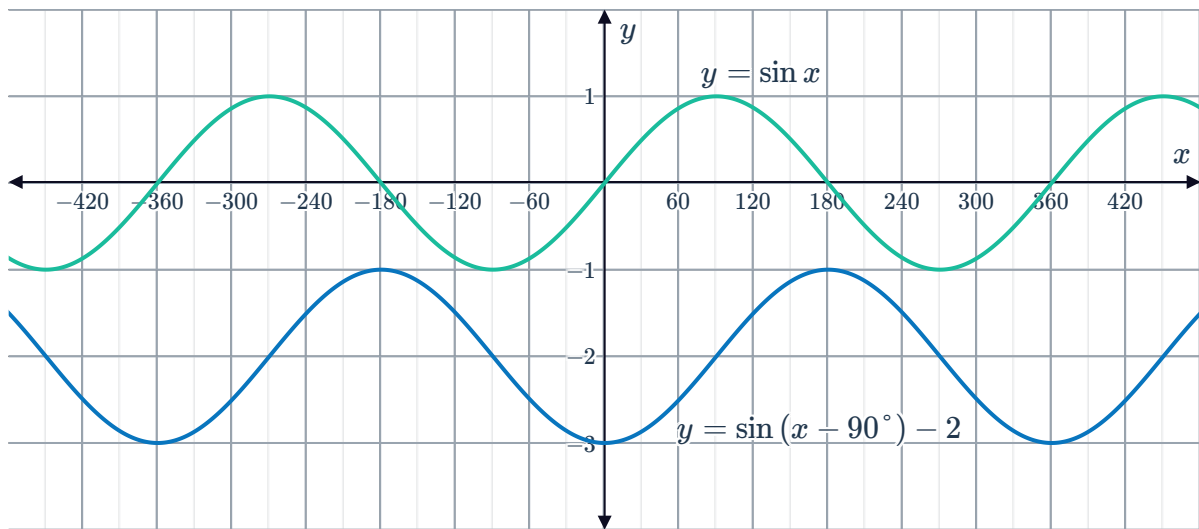




26

a Move the graph to the right by 90° and down by 2 units.

b

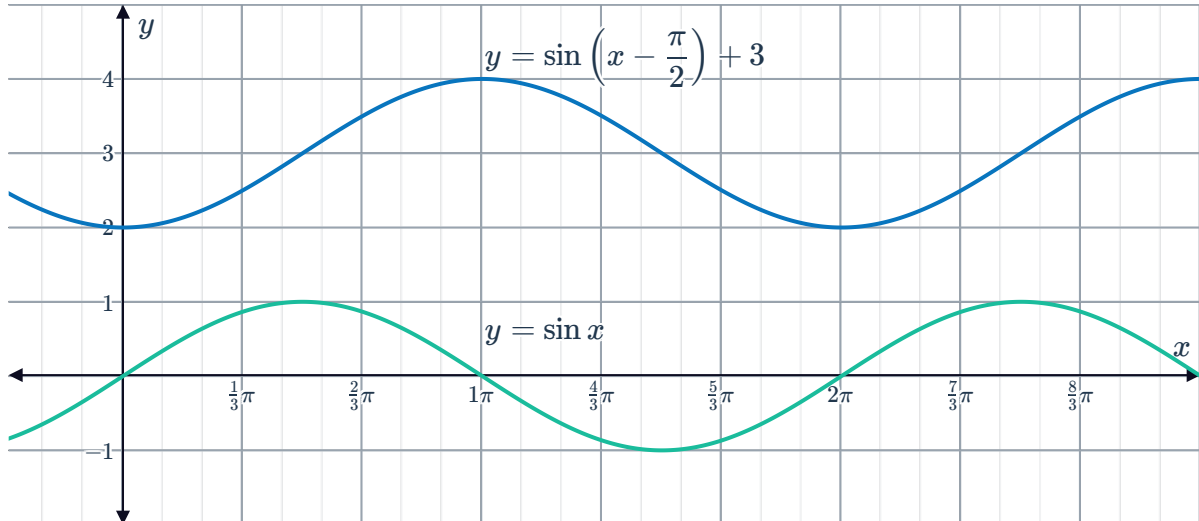


c 360°

27

a Move the graph to the right by $\frac{\pi}{2}$ and up by 3 units.

b



c 2π

28

a $(90^\circ, 5)$

b $(270^\circ, 5)$

c $(90^\circ, 2)$

d $(45^\circ, 1)$

e $(150^\circ, 1)$

f $(90^\circ, 6)$

29

a $\left(\frac{5\pi}{3}, 1\right)$

b $\left(\frac{\pi}{3}, 5\right)$



c $(\pi, 7)$

d $(0, 1)$

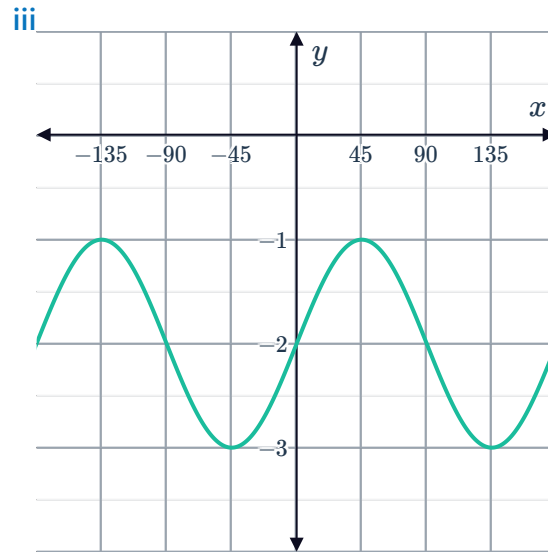
e $(0, 3)$

f $\left(\frac{\pi}{3}, 3\right)$

30

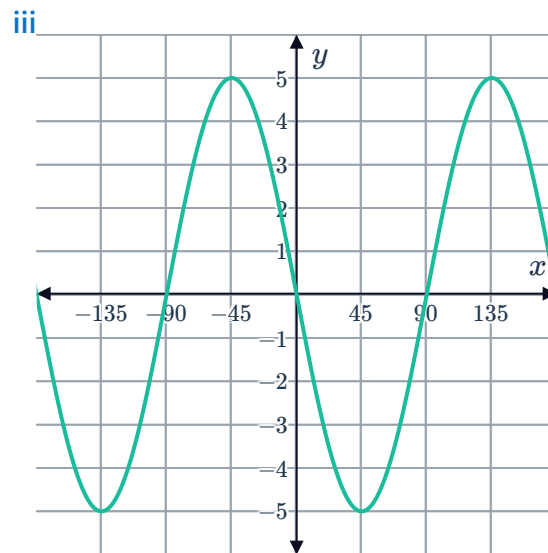
a i $(-\infty, \infty)$

ii $[-3, -1]$



b i $(-\infty, \infty)$

ii $[-5, 5]$



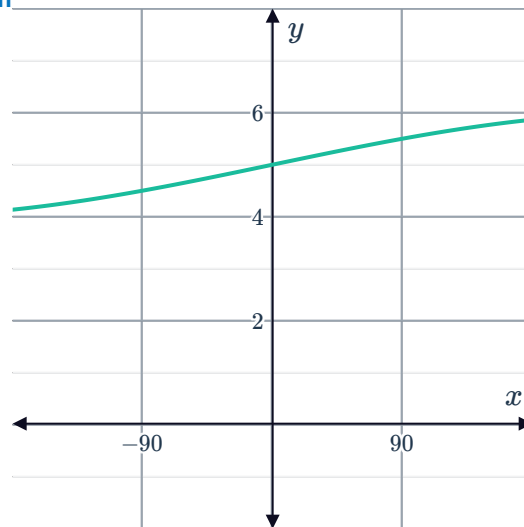
c

i $(-\infty, \infty)$

ii $[4, 6]$



iii

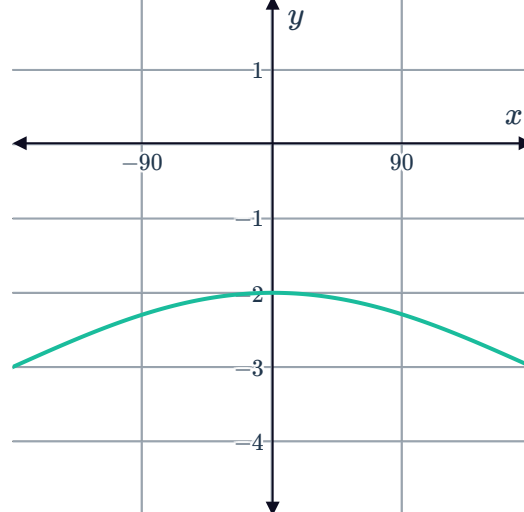


d

i $(-\infty, \infty)$

ii $[-4, -2]$

iii



31

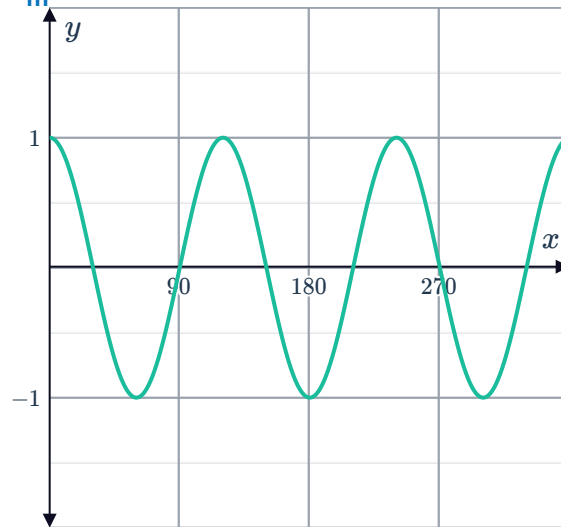
a

i 1

ii 120°



iii

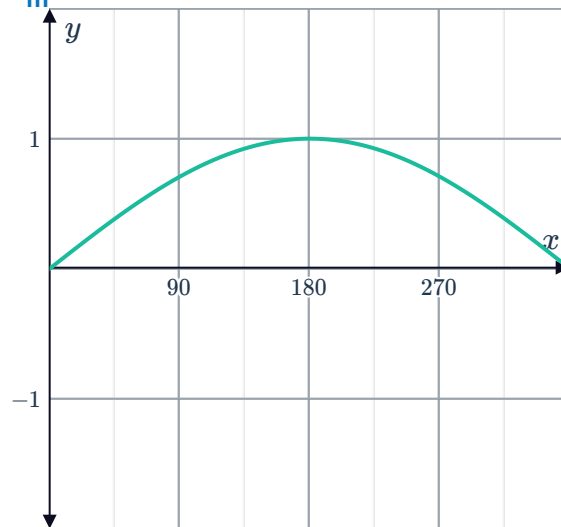


b

i 1

ii 720°

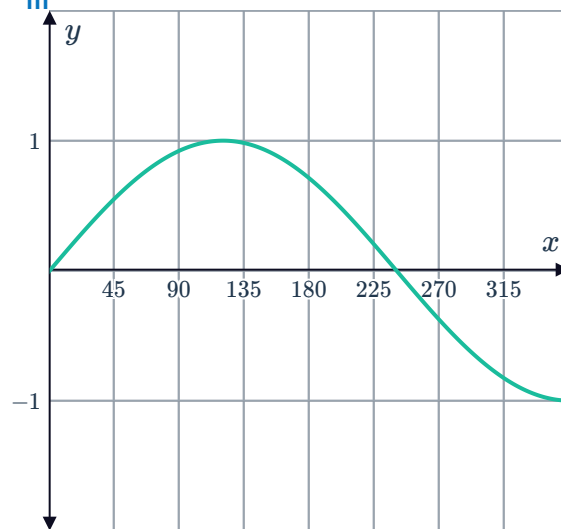
iii



c

i 1

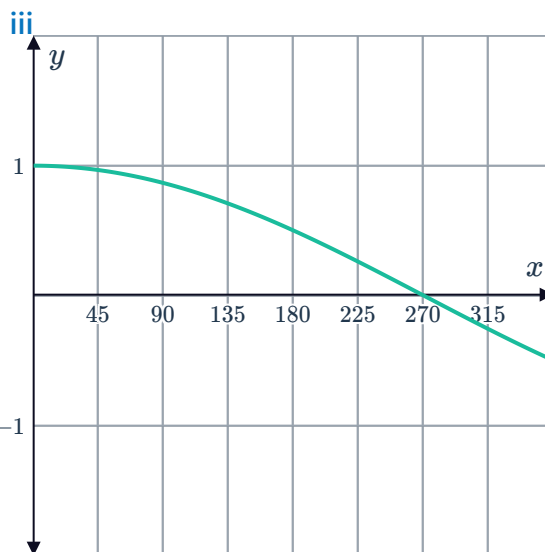
ii 480°



d

i 1

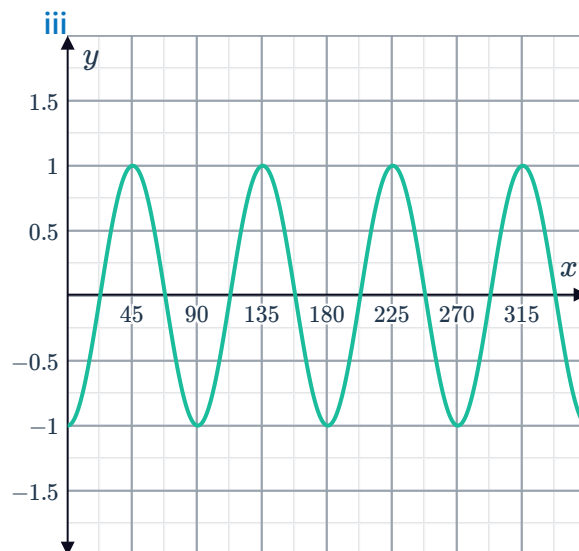
ii 1080°



e

i 1

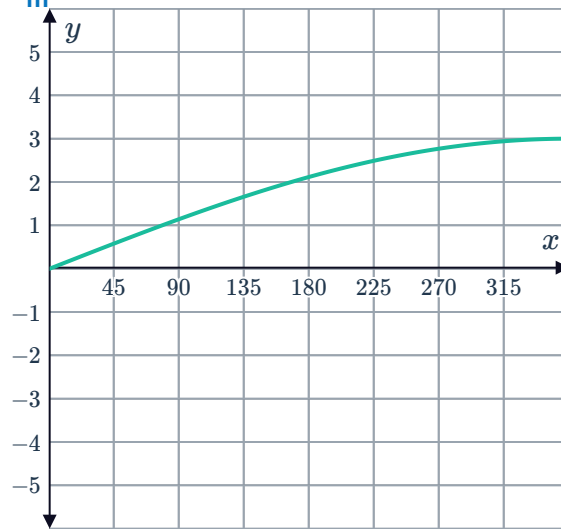
ii 90°



f

i 3

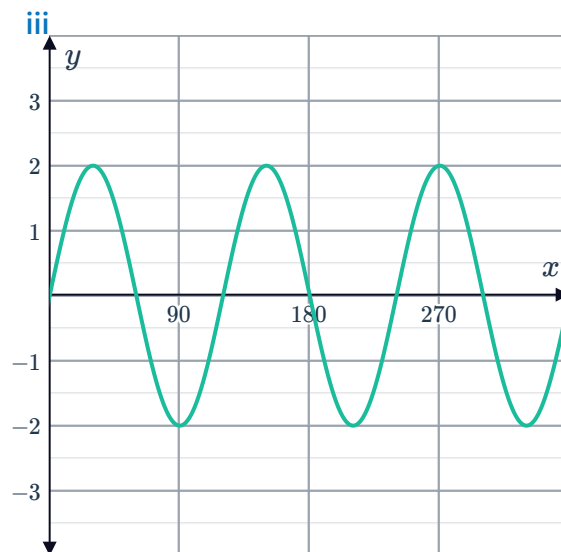
ii 1440°



g

i 2

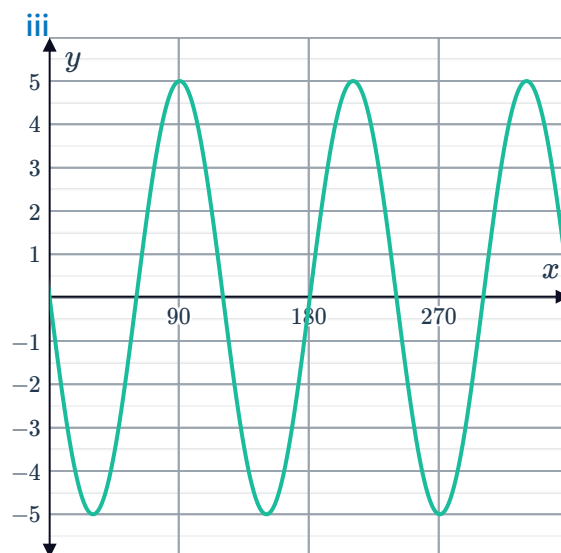
ii 120°



h

i 5

ii 120°





32

a

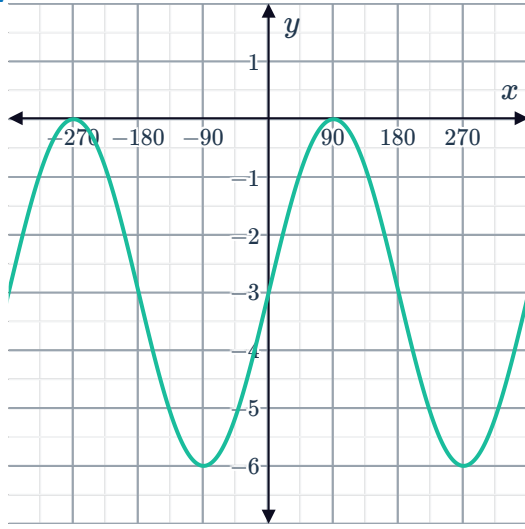
i 360°

ii 3

iii 0

iv -6

v



b

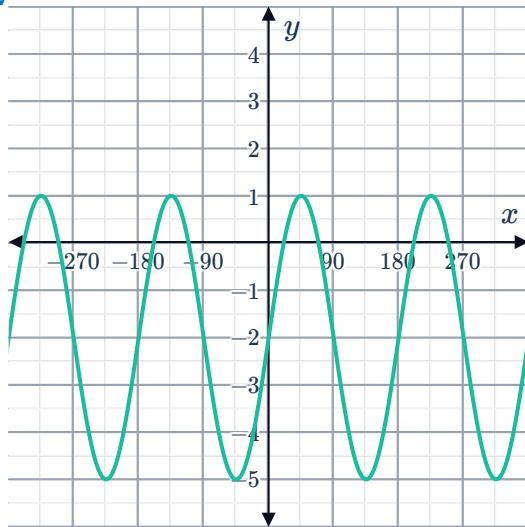
i 180°

ii 3

iii 1

iv -5

v



33

a $y = 3 \sin 2x$

b $y = -\sin 2x$

c $y = 2 \cos 3x$

d $y = -4 \cos \left(\frac{x}{2} \right)$



Applications

34

a $f(t) = 250 + 100 \sin 2\pi t$

c 36 times

e $t = \frac{7}{12}$

b $g(t) = 350 - 100 \sin 4\pi t$

d $\frac{1}{8}$ years

35

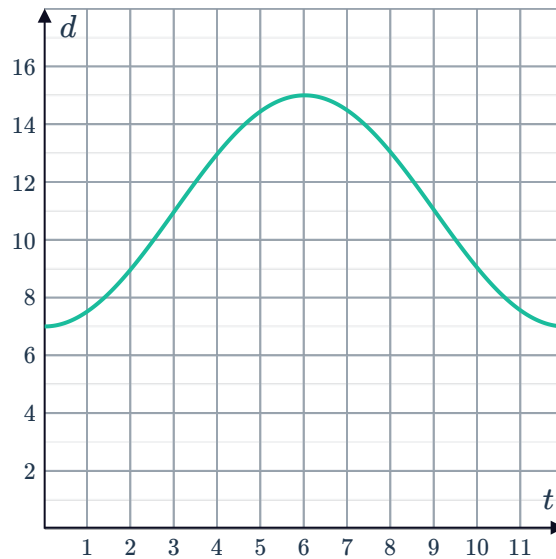
a

Time (t in hours)	0	3	6	9	12
Tide level (d in meters)	7	11	15	11	7

b 4 m

c 12 hours

d



e

i Yes

ii No

iii No

iv No

f $d = -4 \cos \frac{\pi}{6}t + 11$

g 2.66 hours

36

a Sine, the graph's initial value is at the midline.

b $I = \sin \left(\frac{\pi t}{2} \right)$

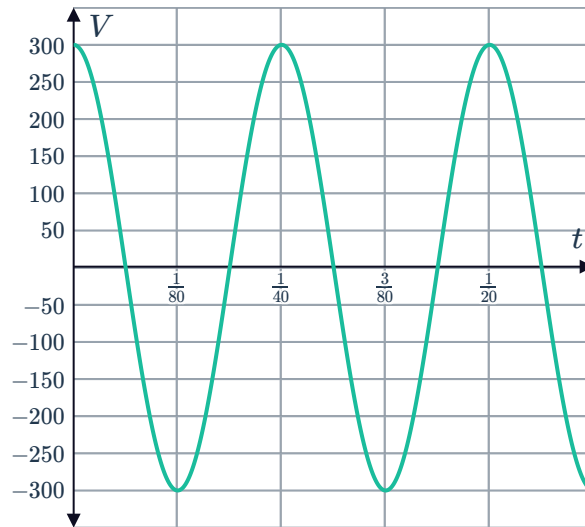


37

- a
- i Yes
 - ii No
 - iii No
 - iv No

b $V = 300 \cos(80\pi t)$

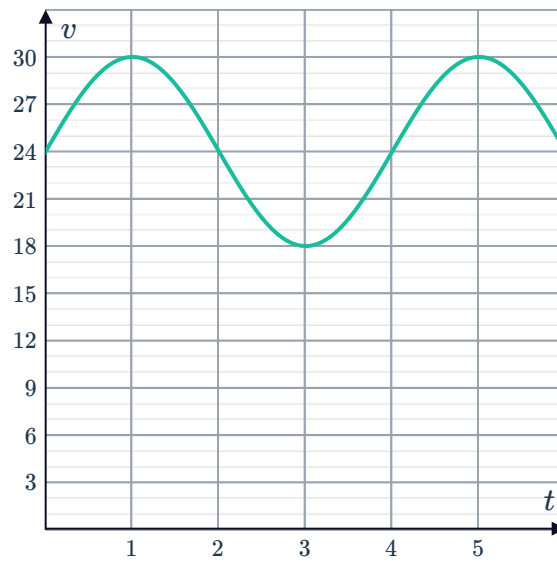
c



- d It would decrease the period of the function and compress the graph horizontally.

38

a



- b 30 units
- c The number of seconds that have passed since the beginning of the section.
- d 24 units